

The Youth Science Journal Of Iraq



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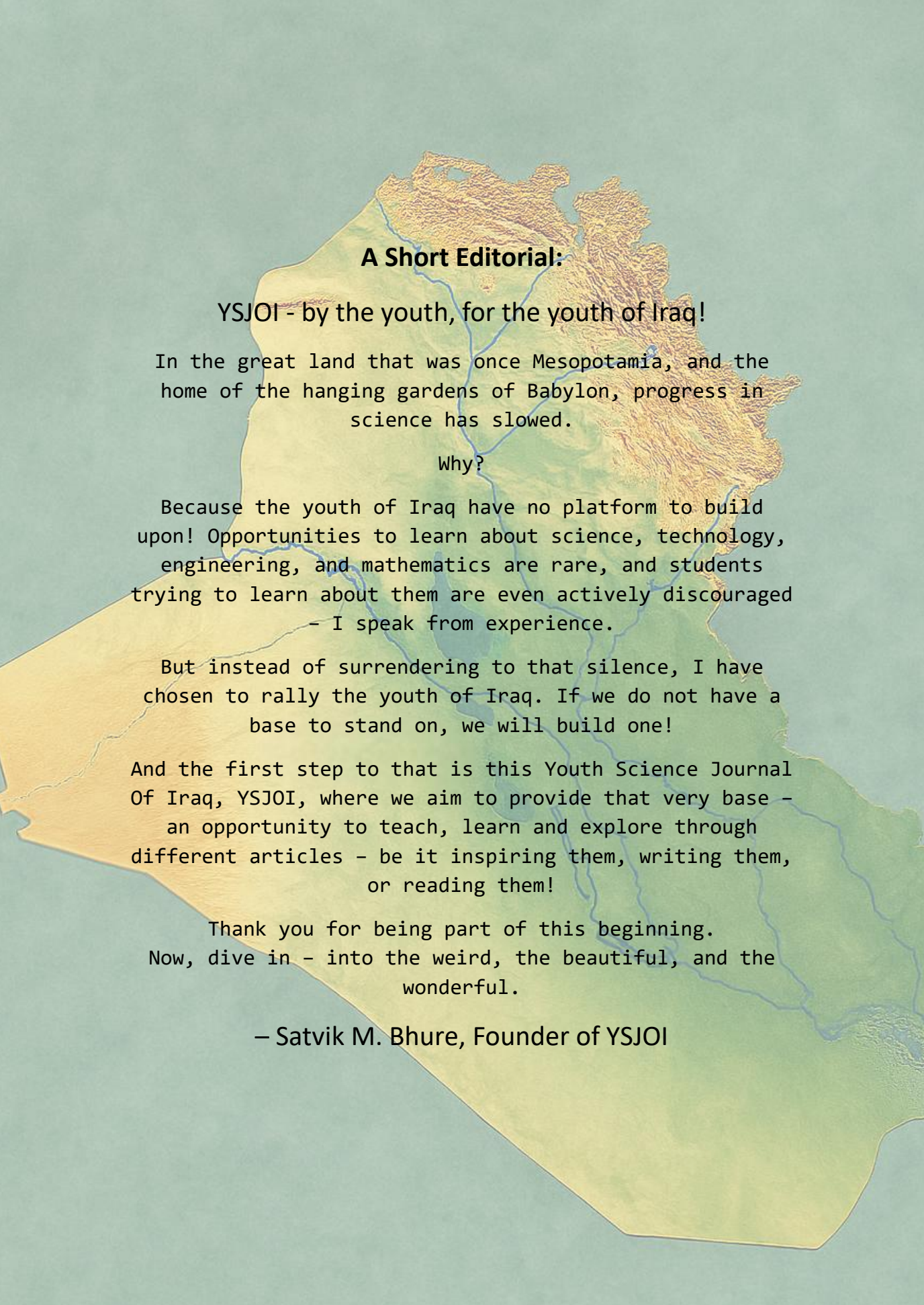
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A Challenge And A Note:

This edition of YSJ0I is our Staff edition, where each of the staff picked out their choice of questions and it was answered. To keep the fun of the edition, the articles may or may not be misattributed to random people; and hence the challenge:

One of the articles of this issue is completely false; our own little April fool's joke. See if you can recognize it!

The answer will be posted in the next issue of YSJ0I!



A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged

- I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base - an opportunity to teach, learn and explore through different articles - be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in - into the weird, the beautiful, and the wonderful.

- Satvik M. Bhure, Founder of YSJOI

Question I: What is the best heat conductor?

— *Safa Wissam*

D for diamond, right? Well, you might just be wrong this time! Diamond has been the best heat conductor for over 250 years... So, why is it not quite the correct answer?

Before the answer is revealed, let's ask another question: What makes a material good at conducting heat?

A substance's heat conductivity depends on its structure; if its internal structure allows energy to move from one atom to another with minimal resistance through carriers such as delocalized electrons in metals, then it's fit for the job.

And that is exactly what diamond is—well, not quite. Diamond is made up of carbon atoms linked by tight bonds in a 3D, tetrahedral structure. Carbon is also a light element, it can vibrate at very high frequencies, allowing heat to zip through the structure. Think of it like a tight guitar string versus a loose one—the tight string transmits energy more efficiently. And since diamond's structure is a repeating lattice, it allows heat to travel in a straight line without facing any obstacles. But, if diamond is made up of carbon,

how can it have delocalized electrons? It doesn't!

Instead of freely-moving electrons, diamond has phonons; a single unit of traveling vibration. Due to diamond's tight, stiff bonds, heat waves (phonons) are able to zip through in an instant!

Recently, scientists at the University of Houston have proven that *Boron Arsenide* is better at conducting heat than diamond—reaching over 2100 W/mK at room temperature, exceeding the theoretically predicted value of just 1280 W/mK. While diamond stands between a range of 2000-2200 W/mK, boron arsenide is a huge competitor, not only physically, but economically as well. It does not require extreme temperatures nor pressures, making it a lot cheaper to produce. Both diamond and boron arsenide can be classified as thermal grades; they efficiently allow extreme heat dissipation in electronics, preventing them from melting under high power. But, here's the kick: boron arsenide is also a great semiconductor. Therefore, not only can it operate as a cooling system, it can also semi-conduct electricity. In conclusion, diamond has its competitor, so it's not exactly the right or wrong answer *yet*—only time will tell.

Question II: What is the universe fated to be?

– Ahmed Muhammed

The Big Bang. The most widely accepted theory about what started this Universe. Since then, it has been expanding at increasing speeds. But how will it end?

There are three main theories: heat death, the Big Rip, and the Big Crush.

First of all, heat death. This is entropy, where all stars eventually stop burning, and all heat undergoes radiation, conduction, and convection, so that the whole world is in thermal equilibrium. In such a case, all energy will be balanced out, and no more reactions, energy transfer, or anything will happen. The Universe would have simply exhausted itself. In this case, the Universe is not destroyed; it just becomes one big beaker at a constant temperature. It's the boring ending.

On the other side, there is the Big Rip. This theory states that the Universe will grow, and continue to grow, accelerating, so that gravity eventually becomes too weak to hold it together, and the Universe rips itself to shreds. It starts with superclusters of galaxies getting

ripped apart, then galaxies breaking down, then solar systems, eventually leading to the breakdown of individual atoms, and everything would get reduced to... nothing. Everything would cease to exist as the four fundamental forces are simply unable to do their job.

The third theory now presents itself: the Big Crush. In this ending, the opposite of the previous one happens.

As the Universe continues to expand, and as it expands violently, gravity gets itself together and pulls in the Universe, fighting against the expansion. Eventually, gravity wins, and the reverse of the Big Bang happens, transforming our Universe from an infinite expanse of free space to a hyper-energized dense cluster of energy, just like how it was when the Universe began. However, this is highly unlikely, as if it does happen, the Universe would simply expand again.

Now this begs the question: if the Big Crush theory is true, then is this the first iteration of the Universe?

Question III: What would happen if the sun disappeared?

– Abdullah Haider

You trust your eyes, don't you? It feels obvious – what you see must be real. But what if your brain isn't actually showing you the world as it is? What if it's guessing... and you're just going along with it?

Scientists now believe the brain works less like a camera and more like a prediction engine. Instead of passively receiving information, it constantly makes rapid guesses about what's out there, using past experiences, memories, and expectations. By the time your eyes send signals, your brain has already built a version of reality – and simply adjusts it if needed.

That explains why optical illusions are so convincing. A still image seems to move. Two identical colours look completely different. Your brain isn't broken – it's just trying to predict too quickly, filling in gaps with assumptions that don't quite match reality.

This doesn't just happen in strange images. It shapes your everyday

life. When you walk into a familiar room, you don't carefully examine every object. Your brain predicts what should be there, saving time and effort. But this shortcut comes with a catch: sometimes, you're not seeing what's actually there – you're seeing what you expect to be there.

Even more unsettling, this means two people can experience the same moment differently. Their brains, shaped by different memories and emotions, construct slightly different realities. So who is right?

Maybe both. Or maybe neither.

The idea is simple, but uncomfortable: reality isn't something you passively observe. It's something your brain actively builds. And if that's true, then the world you trust so completely might not be as solid as it seems. Which begs the question – who CAN you trust?

Question IV: How does the brain dream of things it has never experienced?

– Parwar Phicri

You are surrounded by countless elements, metals and nonmetals alike. Despite most elements being classified as per their properties, a few stand out for their variable behavior under inordinate conditions.

Hydrogen, is just a gas with a single proton and electron, right? Well, not so much. If given the right amount of heat and pressure, metallic hydrogen – a phase of hydrogen where atoms are forced into a metallic state, allowing electrons to move freely – will form. Metallic hydrogen does not exist naturally on Earth – it only exists deep inside giant planets such as Jupiter and Saturn. On Earth, it has only been created in high-temperature, high-pressure laboratory environments; one successful operation was conducted by Harvard University, lasting for just under six months before the experiment was put to an end.

Nitrogen, unlike hydrogen, is a stable gas. When forced under extreme conditions, the molecules break apart, and nitrogen gains

metallic properties, such as thermal stability. One special solid form, Cubic Gauche nitrogen (cg-N), a high-energy-density material (HEDM) that, as the name highlights, stores a huge amount of energy that is released once it decomposes into nitrogen gas. These exotic versions don't last outside high pressure, but they show how versatile nitrogen can be.

Oxygen, on a day-to-day basis, is a familiar diatomic gas that we rely on to live. However, once cooled and compressed, it passes through a series of solid phases with striking colors – red, orange, and deep blue – each reflecting a change in how the O₂ molecules arrange themselves. In some of these dense forms, oxygen becomes electrically conductive, and in its metallic state, it can even act as a superconductor at low temperatures. These transformations don't persist once conditions return to normal, but they reveal how oxygen can shift from a life-sustaining gas into a material with various properties unlike its own.

Question V: How do black holes work?

– *Ali Ahmed Kayani*

Black holes are one of the most mysterious and fascinating objects in the universe. They are regions in space where gravity is so strong that nothing, not even light, can escape. This idea alone sounds almost impossible, yet scientists have proven that black holes are real and exist far beyond our planet.

Most black holes are formed when massive stars collapse at the end of their life cycle. As the star shrinks, its mass gets packed into an incredibly small space, creating a powerful gravitational pull. Anything that comes too close gets pulled in, crossing what scientists call the “event horizon,” a point of no return.

What makes black holes even more interesting is that we cannot see them directly. Since they do not emit light, they are invisible. However, scientists detect them by observing how nearby stars and gases behave. When matter is pulled into a black hole, it heats up and produces intense radiation, giving us clues about its presence.

Despite the progress in space science, many questions about black holes remain unanswered. What happens inside them? Do they lead to another part of the universe? Some theories even suggest the possibility of time distortion, where time slows down near a black hole. These ideas may sound like science fiction, but they are based on real scientific thinking.

At the current moment in time everyone’s guess is good as any other. Maybe there’s magic in a black hole, or maybe another universe. What’s certain is that we will learn more about them in our lifetimes! Maybe it will be one of us, the youth, who figures it out!

Black holes remind us that the universe is far more complex than we imagine. They challenge our understanding of physics and push scientists to explore deeper into space. For students like us, they show that science is not just about answers, but also about endless questions waiting to be explored.

Question VI: Why is the Moon's density so low?

– *Yousif Muhammed*

We have all looked up at the night sky and wondered about the glowing orb that orbits our planet. For centuries, we were told it was a desolate rock, a silent witness to Earth's history made of basalt and regolith. We grew up believing the Apollo missions brought back "moon rocks" that were heavy, solid, and ancient.

But does this actually make sense?

If the Moon were truly a solid sphere of rock, its gravitational pull would be significantly higher than what we observe today. Scientists have recently begun to question the "Lunar Density Paradox." When you look at the way the Moon "rings" like a bell when hit by meteorites—a phenomenon recorded by seismic sensors—it suggests that the interior isn't solid at all. It is porous. It is light. It is, essentially, a celestial ball of aerogel.

Aerogel, often called "frozen smoke," is the lightest solid known to man. It is 99.8% air and an incredible insulator. This explains why the Moon can withstand the

extreme temperature swings of space without crumbling. But how did it get there?

New theories suggest that during the early formation of the solar system, high-pressure "quantum foam" solidified in the cold vacuum, trapping trace amounts of stardust within a silica-based lattice. This created a massive, low-density sphere. The "rocks" brought back by astronauts were actually just the thin, outer crust—a "scab" of space dust that settled on the aerogel core over billions of years.

This leads to an even more exciting realization: if the Moon is a giant ball of aerogel, it doesn't just orbit us; it floats on the solar winds! This is why the magnetic field of the Earth is so important; it keeps our "smoke balloon" from drifting away into the deep dark.

So next time you look up, don't see a heavy rock. See a delicate, frozen cloud of smoke, held in place by the delicate balance of the universe. It really is rather cool to think about, since it is a matter of this balance.

Acknowledgements

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We thank all the scientists who have worked thus far to find all about the world so we can learn about it, and everyone who has supported any member of the YSJOI team in furthering our mission.

Thank you to the rest of the YSJOI team (listed on the next page) for this incredible staff edition and April Fool's edition in the April 2026 issue of YSJOI!

Thanks also to the growing team of YSJOI,
where we have welcomed many new members,
all of who had a hand in this incredible issue
and all the work that went into it:

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Parwar Phicri

Ahmed Muhammed

Safa Wisam

Asal Ibrahim

Syed Minhail Abbas

Abdullah Haider

Yousif Muhammed

Hasar Salah

And we thank you, reader, for supporting our
mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at <https://YSJOI.is-great.org>

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